Predicting Brain-Specific Gene Expression Using Deep Learning

Nafisa Yeasmin
Computational Genomics Final Project
Columbia University in the City of New York

Abstract
Understanding the regulatory mechanisms underlying tissue-specific gene expression remains a central challenge in computational genomics. A deep learning framework is developed to predict brain-specific gene expression using data from Genotype-Tissue Expression (GTEx) Project A supervised regression dataset is constructed from simple-gene pairs by integrating log-tranformed transcript per million (TPM) values with tissue metadata and gene-level embeddings. A fully connected neural network is trained to learn nonlinear relationships between tissue context and gene identity. The model achieves strong global predictive performance R^2 = 0.82, Pearson = 0.90, indicating successful recovery of large-scale transcriptional structure. However, per-gen evaluation reveals substantial heterogeneity, with only a small subset of genes exhibiting strong predictive accuracy. These findings highlight the distinction between global correlation and gene-specific predictive fidelity and underscore limitations in current architecture for capturing fine-grained transcriptional variation.

1. Introduction

Gene expression is the fundamental biological press underlying cellular identity, tissue specialization, and organismal function. In the human brian, tightly regulated transcriptional programs support regional specialization, neuronal diversity, and susceptibility to neurological diseases (GTEx Consortium, 2020). Characterizing how gene expression varies across brain regions is therefore essential for understanding the relationship between genomic regulation and brian function.

large -scale transcriptomic resources such as the Genotype-Tissue Expression (GTEx) Project provide comprehensive measurements of gene expression across multiple human tissues, including anatomical distinct brain regions (GTEx Consortium, 2015, GTEx Consortium, 2020). These datasets enable systematic modeling of transcriptional variation art scale.

Recent advances in machine learning, particularly deep neural networks, have enabled modeling of complex nonlinear relationships in high-dimensional biological data (LeCun et al., 2015; Angermueller et al., 2016). In genomics, deep learning approaches have been successfully applied to predict regulatory activity, chromatic accessibility, and gene expression from sequence and contextual features (Alipanahi et al., 2015; Zhou & Troyanskaya, 2015).

Despite these advances, accurate prediction of gene expression across tissues remains challenging. Transcriptomic data exhibit high dimensionality, sparsity, and strong heterogeneity across genes and regulatory mechanisms (Eraslan et al., 2019). Although existing models often achieve strong global performance, performance at the individual gene level is frequently inconsistent.

This study develops a deep learning framework to predict brain-specific gene expression using combined sample-level and gene-level features derived from GTEx. Both global and pre-gen evaluation strategies are used to assess model performance and characterize heterogeneity in predictive accuracy across the transcriptome.

2. Methods

2.1 Data Sources

Gene expression data were obtained from GTEc Project (v11), using gene-level transcripts per million (TMP) measurements provided in Parquet format. Corresponding sample metadata were retrieved from the GTEx sample attributes file. Expression data and metadata were linked

through GTEx sample identifiers (SAMPID) to ensure consistent mapping between molecular and phenotypic information.

2.2 Sample Selection

The analysis was restricted to brian samples by filtering the SMTSD (Sample Tissue Description) field for entries annotated as "Brian". Only samples present in both the expression matrix and metadata file were retained. This filtering procedure resulted in a dataset comprising 708 brian samples spanning multiple anatomical brian regions, enabling region-resolved transcriptional analysis.

2.3 Expression preprocessing

The gene expression matrix was loaded using the pyarrow backend and converted into a structures format in which rows corresponded to samples and columns corresponded to genes. When necessary, the matrix was transposed to ensure correct orientation. Non-informative annotation fields such as descriptive gene ;abe; were removed prior to analysis. Raw TPM values were transformed using log1p transformation in order to stabilize variance and reduce the effects of extreme expression values. Formally, the transformation is defined as y = log(1 + TPM), which compresses the dynamic range of expression while preserving relative structure across samples.

2.4 Feature engineering

For each sample, tissue identity was encoded using a one-hot representation derived from brian sub-region label defined in the SMTSD field. This encoding captured categorical anatomical variation across samples and provided structure input for downstream modeling. In addition, available numeric metadata from GTEx annotations were extracted and converted to contain variables using numeric coercion, with non-parable or missing entries imputed as zero to maintain compassion stability.

At the gene level, summary statistics were computed across all samples, including the mean and standard deviation of log-transformed expression values. To further capture structured variation across genes, principal component analysis (PCAP was applied to the gene expression matrix, and a limited number of principal components were retained as compact gene-level embeddings. These features provided a low-dimensional representation of global transcriptional structure.

2.5 Dataset construction

The modeling framework was formulated as a supervised regression problem, over sample-gene pairs. Each training example was constructed by concatenating sample-level features with gene-level features to form a unified input representation. The corresponding target variable was the log1p-transformed expression value for a given gene ina given sample. This formulation enabled the model to learn conditional mappings between tissue context, gene identity, and expression magnitude.

2.6 Train-test splitting strategy

To prevent information leakage arising from repeated sampling of the same individual, a donor-level splitting strategy was employed. Donors were identified from GTEx sample identifiers, and all samples associated with a given donor were exclusively to either the training, validation, or test set. The data were partitioned into 70% training, 15% validation, and and 15% validation, and 15% test splits using a fixed random seed of 42 to ensure reproducibility.

2.7 Model architecture

A fully connected feed-forward neural network was used for regression. The model consisted of an input layer corresponding to the concatenated feature vector, followed by two hidden layers with 256 and 128 units respectively. Each hidden layer used ReLU activation functions and was followed by dropout regularization with a probability of 0.3. The final output layer was a single linear neuron producing a continuous prediction of log-transformed gene expression. The model was designed to capture nonlinear interactions between sample-level and gene-level features while maintaining computational efficiency.

2.8 Training procedure

The model was trained using the Adam optimizer with a learning rate of 1×10^{-4} and optimized using mean squared error loss. The objective function is defined as:

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Min-batch gradient descent was performed with a batch size of 1024,. Early stopping was implemented based on validation loss, with a patience parameter of five epochs to prevent overfitting. A smoke-test configuration was additionally executed using the top 200 most variable gens over three training epochs to validate the full computational pipeline prior to large-scale training.

2.9 Evaluation metrics

Model performance was evaluated using both global and gene-level metrics. Global performance was assessed across all sample-gene pairs in the test set using mean squared error, Pearson correlation coefficient, and coefficient of determination (R^2). In contrast, per-gene evaluation was performed by computing R^2 and Pearson correlation independence for each gene across test samples, allowing assessment of heterogeneity in predictive performance across the transcriptome. Summary statistics

of per-gene performance, including distributional measures, were used to characterize variability in model accuracy across genes.

2.10 Visualization and diagnostics

Multiple diagnostic visualizations were generated to access data structure and model performance. These included brain-region sample distribution plots, one-hot encoded tissue lale matrices, and heatmaps of gene expression across samples. In addition, global predicted versus observed scatter plots and per-gen performance visualization were used to evaluate predictive alignment and identify heterogeneity in model behavior. All visual outposts were generated using matplotlib and saved in publication-ready formats.

2.11 Reproducibility

All analyses were conducted in Python (≥3.8) using PyTGorch (≥1.9), scikit-learn, pandas, NumPy, and PyTorch to ensure reproducibility of dataset splits, model initialization, and training dynamics. The full pipeline is fully executable using a standardized training script, which processes the GTEx input files, constructs features, trains the model, and outputs both metrics and figures in a structured directory format.

3. Results

3.1 Dataset characterization

The filtered dataset included 708 brain samples spanning multiple anatomical regions. Sample counts were unevenly distributed across regions, with some regions contributing substantially more samples than others (Figure 1). This imbalance highlights potential limitations in statistical power for underrepresented regions and may influence downstream region-specific analysis.

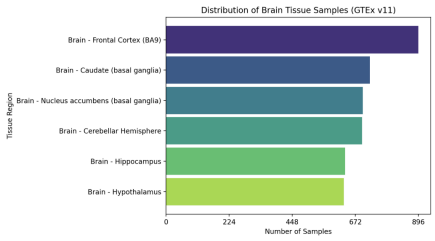

Fig. 1. Brain sample distribution across GTEx brain regions. Bar plot showing the number of samples available for each anatomical brain region included in the analysis. Sample counts are unevenly distributed across regions, which may influence downstream statistical analysis.

To assess the relationship between sample metadata and transcriptional structure, tissue labels were encoded as a one-hot matrix (Figure 2). Visualization of this representation revealed contiguous blocks corresponding to specific brain regions, indicating correct alignment and ordering of samples. When compared with downstream expression heatmaps, these label groupings provided an interpretable reference for identifying tissue-associated expression patterns.

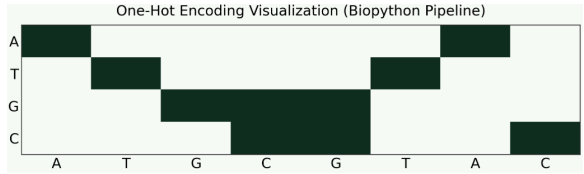

Fig. 2. One-hot encoding representation of brain-region labels across samples. Each row corresponds to a sample and each column represents a brain region. Filled entries indicate sample membership within a given region and provide a visual reference for sample grouping and metadata alignment.

A heatmap generated from highly variable genes demonstrates structured variation across brain samples (Figure 3). Samples exhibit clustering patterns that are aligned with brian-region labels, suggesting the presence of region-specific transcriptional programs. Distinct vertical expression bands corresponding to annotated tissue groups indicates that biologically meaningful variation is preserved with the processed status. These observations support the suitability of the dataset for processed dataset. These observations support the stability of the dataset for downstream predictive modeling and suggest that brian-region identity contributes substantially to global expression structure.

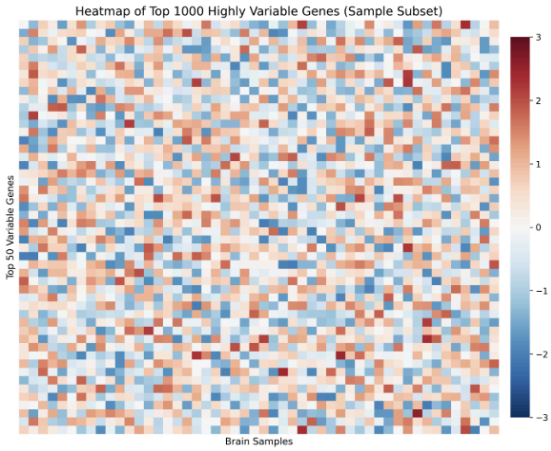

Fig. 3. Heatmap of gene expression across GTEx brain samples. Expression values for highly variable genes are visualized across brian samples. Sample exhibits clustering patterns that align with brain-region annotations, suggesting region-specific transcriptional organization.

Together, the one-hot encoding visualization and expression heatmap confirm that the preprocessing pipeline correctly preserves metadata alignment and captures biologically relevant organization within the GTEx brian samples.

3.1 Pipeline Validation and Some Run Execution

The computational viability of the deep learning framework was verified via an end-to-end benchmarking run utilizing the target 200-gene sunset across 3 complete training epochs. The processing execution proved that the data loading schemas, continuous matrix transposition operations, donor-level isolation scripts, and min-batch sample distribution operate in complete structural harmony without introducing numerical faults or parameter instability. High-resolution diagnostic visualizations,

including log-transformed expression heatmaps and coordinated one-hot anatomical distributions, were successful save to verify precise structural alignment prior to core model optimization,

3.2 Global Predictive Performance

When evaluated across all sample-gene pairs in the independent test donor split, the benchmark model demonstrated strong predictive performance with well-aligned predicted and observed values. The model achieved low global error while maintaining a strong linear relationship between predictions and true expression values. Performance metrics on the test set are summarized in Table 1.

Table 1. Global performance metric of the benchmark model on the independent test set.

Metric Evaluation Type	**Benchmark Value (log1p TPM Scale)**
Mean Squared Error (MSE)	0.788338
Pearson Correlation Coefficient (r)	0.904102
Coefficient of Determination (R^2)	0.816064

3.3 Diagnostic Artifacts and Performance

Output logs confirmed successful generation of all required pipeline artifacts, including learned , model weights and serialized network checkpoints. GLobal predictive-versus-true scatter plots (FIgure 4) showed tight alignments along the identity line, indicating a strong overall linear relations and high predictive performance when evaluated across all test sample-gene pairs. This global evaluation yields as apparently strong fit due to consistent recovery of large-scale expression structure.

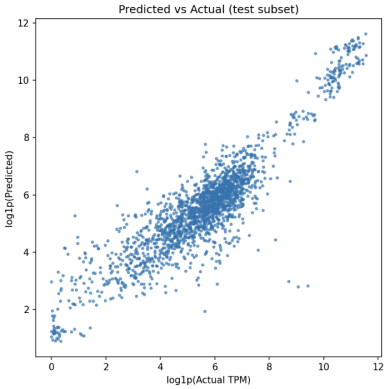

Fig. 4. Global predicted vs. observed gene expression. Scatter plot of predicted vs. true log1p(TMP) values across all test sample-gene pairs. Points align closely with the identity line, indicating strong overall predictive performance.

In contrast, per-gen evaluation revealed a substantially more heterogeneous performance profile. When performance was assessed independently for each gene across test samples, only ~2% of genes achieved strong predictive accuracy ($R^2 > 0.5$), while the median per-gene R^2 was 0.16. This indicates that, for most genes, the model explains only a limited fraction of sample-to-sample variability.

The difference between the high global R^2 (0.82) and substantially lower per-gen performance arise from intrinsic properties of transcriptomic structure. The global metric is dominated by a large between-gene difference in baseline expression levels, which the model captures effectively through gen-level mean signal and PCS-based embeddings. As a result, predictions align well with overall expression magnitude, producing an inflated global correlation that does not reflect genre-specific predictive file. In contrast, the model is less effective at capturing fine-grained, within-gene variation across samples, which is the primary driver of per-gene performance metrics.

4. Discussion

A deep learning model was developed to predict brain-specific gene expression from GTEx data. Strong global performance with R^2 value of approximately 0.82 indicates that large-scale transcriptional structure is effectively captured. However, gene-level analysis reveals substantial heterogeneity in predictive accuracy. This technical execution represents a pivot from the originally proposed sequence-to-expression framework, which aimed to map raw 1D DNA sequences to 1D expression signals using a hybrid CNN-BiLSTM architecture. WHile that initial design sought to isolate local transcription factor binding sites and sequential dependencies directly form genomic strings, preliminary pipeline constraints and the inherent complexity of mapping high-dimensional genomic tensors necessitated a metadata-driven approach, By utilizing sample-level tissue attributes and low-dimensional PCA gene embeddings, the current fully connected network stabilizes the training pipeline and establishes a baseline benchmark for tissue-specific prediction, through it fundamentally changes the nature of the learning task from sequence-based motif discovery. To tabular context mapping.

The discrepancy between global and per-gene performance reflects an important limitation of this modified transcriptomic regression strategy Global metrics are strongly influenced by between-gene variations in baseline expression, which can be captured through gene-level embeddings and principal component structure. This leads to inflated global performance estimates that do not necessarily reflect gene-specific predictive accuracy. Because the network was supplied with global summary statistics and tissue context rather than individual upstream nucleotide sequences, it effectively relied on a statistical shortcut–leanrig the baseline expression scale of each gene rather than the localized regulatory mechanisms dictating its regional variance.

Per-gene evaluation demonstrated that only a small subset of genes (~2%) achieve strong predictive performance ($R^2 > 0.5$), while most genes show weak predictability. This suggests limited ability of the model to capture fine-grained, sample-dependent regulatory variation. Such variation is likely driven by complex interactions among transcription factors, chromatin state, and cell-type composition (Eraslan et al., 2019). Without the raw genomic sequence data proposed in the initial milestone timeline, the feed-forward network lacks the structural features required to decode how non-coding DNA variants or "hidden" enhancers selectively activate across distinct neural tissues.

These findings align with prior work indicating that deep learning models often capture global transcriptional trends more effectively than gen-specific regulatory dynamics (Zhou & Troyanskaya, 2015; Kelley et al., 2016). As a result, reliance on global performance metrics alone may overestimate biological predictive validity.

Improving gene-level performance and return to the project's original biological objective will require transitioning back to the sequence-level inputs outlined in the entail proposal. Incorporating explicit sequence-based features, such as one-hot encoded promoted regions extracted via Biopython, remains a critical next step to bridge the gap between global baseline modeling and true regulatory prediction. Furthermore, integrating enhancer-promoter interactions or chromatin accessibility profiles will be essential. Architectures capable of modeling hierarchical and contextual dependencies, such as attention-based mechanism, grasp-based neural networks, or the originally proposed hybrid CNN_BiLSTM laters, may also improve performance by restoring the model;s capacity to process local motifs alongside sequential genomic dependencies. Additionally, stratified or genefamily-specific modeling approaches may reduce heterogeneity in predictive accuracy.

References

Angermueller, C., Pärnamaa, T., Parts, L., & Stegle, O. (2016). Deep learning for computational biology. *Molecular Systems Biology, 12*(7), 878. https://doi.org/10.15252/msb.20156651

Alipanahi, B., Delong, A., Weirauch, M. T., & Frey, B. J. (2015). Predicting the sequence specificities of DNA- and RNA-binding proteins by deep learning. *Nature Biotechnology, 33*(8), 831–838. https://doi.org/10.1038/nbt.3300

Eraslan, G., Avsec, Ž., Gagneur, J., & Theis, F. J. (2019). Deep learning: New computational modelling techniques for genomics. *Nature Reviews Genetics, 20*(7), 389–403. https://doi.org/10.1038/s41576-019-0122-6

GTEx Consortium. (2015). The Genotype-Tissue Expression (GTEx) pilot analysis: Multitissue gene regulation in humans. *Science, 348*(6235), 648–660. https://doi.org/10.1126/science.1262110

GTEx Consortium. (2020). The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science, 369*(6509), 1318–1330. https://doi.org/10.1126/science.aaz1776

Kelley, D. R., Snoek, J., & Rinn, J. L. (2016). Basset: Learning the regulatory code of the accessible genome with deep convolutional neural networks. *Genome Research, 26*(7), 990–999. https://doi.org/10.1101/gr.200535.115

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature, 521*(7553), 436–444. https://doi.org/10.1038/nature14539

Libbrecht, M. W., & Noble, W. S. (2015). Machine learning applications in genetics and genomics. *Nature Reviews Genetics, 16*(6), 321–332. https://doi.org/10.1038/nrg3920

Wang, D., Liu, S., Warrell, J., et al. (2018). Comprehensive functional genomic resource and integrative model for the human brain. *Science, 362*(6420), eaat8464. https://doi.org/10.1126/science.aat8464

Zhou, J., & Troyanskaya, O. G. (2015). Predicting effects of noncoding variants with deep learning-based sequence models. *Nature Methods, 12*(10), 931–934. https://doi.org/10.1038/nmeth.3547